AI Developer Assessment Task

AI-Powered Tea Packaging Optimization Platform

Company: Innovacio Technologies Pvt. Ltd.
Position: AI Engineer / AI & Backend Developer
Difficulty Level: Intermediate to Advanced
Estimated Time: 1 Working Days
Submission: GitHub Repository + Documentation + Demo Video

Overview

The objective of this assessment is to evaluate the candidate's ability to:

- Analyze a real-world business problem
- Design an AI-assisted optimization system
- Build scalable backend APIs
- Apply optimization algorithms
- Structure clean, maintainable code
- Design databases
- Develop production-ready software

The project simulates a real client requirement from the tea export industry.

Business Problem

Tea manufacturers export thousands of cartons worldwide.
Today, packaging dimensions, carton sizes, pallet layouts, and container loading are manually decided, resulting in:

- Low container utilization
- High freight costs
- Packaging waste
- Increased material costs
- Poor space optimization

The goal is to build an AI-based system that recommends the optimal packaging configuration.

Objective

Develop a web application where the user provides:

- Tea Density
- Package Weight
- Shipment Quantity

The system should automatically recommend:

- Inner Pouch Dimensions
- Master Carton Dimensions
- Units Per Carton
- Pallet Configuration
- Best Container Type
- Cost Optimization
- Current vs AI Comparison

Functional Requirements

Module 1 — Dashboard

Display

- Total Simulations
- Total Savings
- Average Container Utilization
- Recent Simulations

Module 2 — New Optimization

User Inputs

Field	Type
Tea Density	Number
Package Weight	Dropdown
Shipment Quantity	Number
Shipment Type	Total Weight / Per Container
Package Shape	Square / Round

Packaging Material	Paper / Plastic / Metal
Target Market	Optional

Module 3 — AI Package Recommendation

The system should calculate

- Product Volume
- Suggested Package Dimensions
- Fill Ratio
- Material Usage

Display

- Best Package
- Alternative Package Options
- Estimated Cost

Module 4 — Carton Optimization

Calculate

- Carton Length
- Carton Width
- Carton Height
- Units Per Carton
- Carton Weight
- Board Grade

Module 5 — Pallet Optimization

Calculate

- Cartons Per Layer
- Layers
- Cartons Per Pallet
- Pallet Height
- Total Weight

Module 6 — Container Optimization

Compare

- 20 GP

- 40 GP
- 40 HC

Calculate

- Container Utilization
- Empty Space
- Cartons Per Container
- Total Units
- Freight Cost

Module 7 — Current vs AI Comparison

Create a dashboard comparing

Parameter	Current	AI	Improvement
Package Size	✓	✓	%
Carton Size	✓	✓	%
Units Per Carton	✓	✓	%
Cartons Per Pallet	✓	✓	%
Containers Required	✓	✓	%
Packaging Cost	✓	✓	%
Freight Cost	✓	✓	%
Container Utilization	✓	✓	%
Total Cost	✓	✓	%

AI Logic

The solution should use
Tea Density

↓

Calculate Volume

↓

Generate Package Options

↓

Optimize Carton

↓

Optimize Pallet

↓

Optimize Container

↓

Compare Multiple Scenarios

↓

Recommend Lowest Cost Solution

Candidates may use:

- Mathematical formulas
- Rule-based optimization
- Google OR-Tools
- SciPy Optimization
- Custom algorithms

Machine Learning is optional.

Technical Requirements

Frontend

Preferred

- React.js
- Next.js
- TypeScript

Backend

Preferred

- Python
- FastAPI

Database

Any relational database
Preferred

- PostgreSQL

API

REST API
Minimum endpoints
POST /simulation

GET /simulation

GET /simulation/{id}

POST /compare

POST /optimize/package

POST /optimize/carton

POST /optimize/pallet

POST /optimize/container

Database Design

Create tables for

- Users
- Simulations
- Tea Density
- Packaging Materials

- Package Types
- Cartons
- Pallets
- Containers
- Results

UI Requirements

Create modern SaaS UI
Pages

- Dashboard
- New Simulation
- Results
- Current vs AI Comparison
- History

Use

- Tailwind
- Material UI
- Shadcn UI

Bonus Features

Additional points for:

- Authentication
- Docker
- Swagger Documentation
- Unit Tests
- CI/CD
- AI Chat Assistant
- 3D Container Visualization
- Export to PDF
- Export to Excel
- Responsive Design

Deliverables

The candidate should submit:

Source Code

GitHub Repository

Documentation

README containing
- Setup Instructions
- Architecture
- Folder Structure
- Assumptions
- API Documentation

Demo

5–10 minute video explaining
- Architecture
- AI Logic
- Application Demo

Evaluation Criteria

Criteria	**Weight**
Code Quality	20%
Backend Architecture	15%
AI / Optimization Logic	25%
API Design	10%
Database Design	10%
UI/UX	10%
Documentation	5%
Problem Solving	5%

Submission Checklist

- GitHub Repository
- README.md
- Database Schema
- API Documentation (Swagger/OpenAPI)
- Setup Guide
- Demo Video
- Screenshots of the application

Assessment Guidelines

- You may use AI coding assistants (e.g., GitHub Copilot, ChatGPT, Claude) to accelerate development, but you must understand and be able to explain all generated code during the technical interview.
- Focus on clean architecture, modular code, and maintainability rather than implementing every possible feature.
- Clearly document any assumptions made due to incomplete business requirements.
- Production-quality coding practices, error handling, input validation, logging, and API documentation are expected.
- The optimization logic should be transparent and explainable, with comments or documentation describing how recommendations are generated.

Interview Follow-up

Candidates who successfully complete the assignment will participate in a 90-minute technical interview, where they will be asked to:

- Walk through their solution architecture.
- Explain their optimization approach and trade-offs.
- Demonstrate the application live.
- Discuss how they would scale the platform for enterprise use.
- Answer questions about the code they submitted.

This assignment is designed to evaluate not only coding ability but also software architecture, optimization thinking, documentation quality, and the ability to solve a real-world business problem.